

MATH 381 Exam 2

March 13, 2026

Name (as on Gradescope): Solutions

Student #: _____

Problem:	1	2	3	Total
Points:	10	5	15	30

INSTRUCTIONS:

- You have 50 minutes to take the test.
- Write your solution below the problem. There is scratch paper at the back of the test.
- If you need extra space to write your solution, use the paper at the back and indicate on which page your solution continues.
- The test is double-sided. Make sure you are reading the backs of pages!
- Unless otherwise stated, show all your work for full credit.
- You are allowed to use one 8.5"×11" sheet of notes, front and back.
- Calculators are not allowed.

Good luck!

1. (10 points) A certain radio station plays two types of songs, hip-hop songs and jazz songs. Every song played is exactly one of these two types.

- If the station plays a hip-hop song, the probability the next song will be **hip-hop** is 60%.
- If the station plays a jazz song, the probability the next song will be **jazz** is 30%.

In the long run, what fraction of the station's songs will be hip-hop songs?

$$P = \begin{matrix} & \begin{matrix} H & J \end{matrix} \\ \begin{matrix} H \\ J \end{matrix} & \begin{pmatrix} 0.6 & 0.4 \\ 0.7 & 0.3 \end{pmatrix} \end{matrix}$$

$$I - P = \begin{pmatrix} 0.4 & -0.4 \\ -0.7 & 0.7 \end{pmatrix}$$

$$(\pi_1, \pi_2) \begin{pmatrix} 0.4 & -0.4 \\ -0.7 & 0.7 \end{pmatrix} = 0 \quad \pi_1 + \pi_2 = 1$$

$$0.4\pi_1 - 0.7\pi_2 = 0$$

$$\pi_1 + \pi_2 = 1$$

$$1.1\pi_1 = 0.7$$

$$\boxed{\pi_1 = \frac{7}{11}}$$

2. (5 points) Suppose we take 100 independent random samples from the same probability distribution f . The sample average of these samples is 60, and the sample standard deviation is 15. Fill in the blanks in the following sentence:

Assuming that we have taken enough samples for the central limit theorem to be relevant, we can be about 95% confident that the true mean of f is between 57 and 63.

The second two blanks should be specific numbers. Show the work you used to get these numbers.

$$\frac{2\sigma}{\sqrt{n}} = \frac{2 \cdot 15}{\sqrt{100}} = 3$$

$$60 \pm 3$$

3. Two people play the following game. There are 20 balls, each labeled with a different integer from 1 to 20. The balls are divided between the two players, with Player 1 starting with balls 1 through 10 and Player 2 starting with balls 11 through 20. Each turn, a fair 20-sided die is rolled. If the die rolls the number k , then the person holding ball k has to give that ball to the other player. Let X_t be the number of balls that Player 1 has after t turns (so $X_0 = 10$). Then $X_0, X_1, X_2, \dots$ is a Markov chain.

(a) (3 points) What is the probability that $X_2 = 12$?

$$\begin{aligned}
 P(X_2=12) &= P(X_1=11 \mid X_0=10) P(X_2=12 \mid X_1=11) \\
 &= \frac{10}{20} \cdot \frac{9}{20} \\
 &= \boxed{\frac{9}{40}}
 \end{aligned}$$

(b) (2 points) Is this chain ergodic? Explain your answer.

No. Because X_t changes by 1 or -1 each turn, X_t is always even for even t and odd for odd t . Therefore, for a state to return to itself, the chain must take an even number of steps.

It is not enough to say the chain has no self loops, or that it takes at least two steps for a state to return to itself. For example, the following chain has no loop but is still ergodic:



- (c) (10 points) Suppose we want to modify the previous Markov chain so that the stationary distribution is proportional to the function

$$F(x) = x(20 - x).$$

Here, x is a state (so it is an integer from 0 to 20). Complete the following code to simulate a Metropolis chain with the chain on the previous page as the proposal chain, and F as the weight function. The sample collected at each step of the simulation is the state itself, and at the end the average of the samples is printed.

```
import random

def step(x):    # Returns next state of chain, if current state is x
    k = random.randint(1,20)    # Uniform random integer from 1 to 20
    if k <= x:
        y = x-1                # y is the proposed next state
        P_xy = x/20             # Transition probability from x to y
        P_yx = (20-y)/20       # Transition probability from y to x
    else:
        y = x+1
        P_xy = (20-x)/20
        P_yx = y/20

    F_old = x*(20-x)
    F_new = y*(20-y)
    alpha = (F_new / F_old) * (P_yx / P_xy)

    r = random.random()        # Uniform random real number in [0,1)
    if r < alpha:
        return y
    return x

def simulate(sample_size=1000000, burn=10000):
    state = 10
    for t in range(burn)
        state = step(state)

    samples = []
    for t in range(sample_size)
        state = step(state)
        samples.append(state)

    return sum(samples) / len(samples)

print(simulate())
```


